BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: PR XI (001) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: Instructions (printed in Urdu on the sheet): 1. Each question has four circles; fill in only the circle of the correct answer. 2. Use a blue or black pen to fill the circles. 3. Filling more than one circle will be treated as a wrong answer.

Q1. Enthalpy means total heat content of a system at constant:

(A) Temperature
(B) Pressure
(C) Volume
(D) Moles

Q2. As the number of effective collisions between the reacting particles increase, the rate of reaction.....

(A) Decreases
(B) Increases
(C) Remains the same
(D) Depresses

Q3. If mole fraction one component of a solution is 0.3, then the mole fraction of second component of that solution will be.....

(A) 0.3
(B) Zero
(C) 1
(D) 0.7

Q4. The molal freezing point constant of a solution is called...

(A) Cryoscopic constant
(B) Ebullioscopic constant
(C) Gas constant
(D) Absolute zero

Q5. Which of the given is not a state function?

(A) Enthalpy
(B) Heat
(C) Temperature
(D) Pressure

Q6. The oxidation number of Cl in $HClO_4$ is.....

(A) +3
(B) +5
(C) +7
(D) +1

Q7. Greater number of hydrogen bonds are found in.....

(A) HF
(B) H_2O
(C) NH_3
(D) CH_4

Q8. London forces are strongest in.....

(A) F_2
(B) Cl_2
(C) Br_2
(D) I_2

Q9. KNO_3 and $NaNO_3$ both have same Rhombohedral Crystal structure this phenomenon is...

(A) Allotropy
(B) Polymorphism
(C) Anisotropy
(D) Isomorphism

Q10. Kc > Kp if Δn is.....

(A) Zero
(B) +ve
(C) −ve
(D) None of these

Q11. The unit of Kw is.....

(A) $mol.dm^3$
(B) $mol^2.dm^{-6}$
(C) $mol^{-2}.dm^{-6}$
(D) $mol^2.dm^{-3}$

Q12. For acidic solutions.....

(A) pH < 7
(B) pH > 7
(C) pH = 7
(D) pH ≥ 7

Q13. What will be the value of temperature and pressure at standard conditions i.e. STP?

(A) 100°C, 1 atm
(B) 298 k, 1 atm
(C) 273 k, 760 mmHg
(D) 0°C, 76 cm Hg

Q14. Bohr's theory cannot be applied on:

(A) H^{+1}
(B) He^{+1}
(C) Li^{+2}
(D) B^{+3}

Q15. Total number of orbitals in "M" shell are.....

(A) 2
(B) 4
(C) 5
(D) 9

Q16. The bond angle in NH_3 is.....

(A) 90°
(B) 104.5°
(C) 107.5°
(D) 109.5°

Q17. Which one of the given will have linear geometry?

(A) $BeCl_2$
(B) H_2O
(C) CH_4
(D) BF_3

Q18. The value of ideal gas constant is.....$atm.dm^3.mol^{-1}.K^{-1}$

(A) 0.0821
(B) 1.98722
(C) 62.364
(D) 8.3143

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Answer any TEN parts. Each part carries FOUR marks.

1. What is limiting reactant? How will you identify a limiting reactant?

2. Apply the Hund's rule for arrangement of electrons in equal energy orbitals of carbon and nitrogen elements.

3. Indicate the type of overlapping of orbitals which result in the formation of sigma (δ) Pi(π) bond.

4. Describe the Dalton's law of partial pressure. Is this law applicable for reacting gases?

5. Why is food cooked rapidly in pressure cooker?

6. Differentiate between isomorphism and polymorphism.

7. Derive a relationship between Kp and Kc.

8. How a buffer solution resist a change in pH?

9. Illustrate the importance of orientation of colliding molecules for a reaction to occur.

10. Prove that the sum of mole fractions of two components of a solution is equal to one.

11. Develop a relationship between heat supplied at constant pressure (qp) and change in enthalpy (ΔH).

12. Why is $BeCl_2$ linear in shape while $SnCl_2$ possess angular geometry?

13. Balance the following equations by half reaction method. (a) $Sn^{2+} + I_2 \rightarrow Sn^{4+} + I^-$ (Neutral medium) (b) $Zn + Cr_2O_7^{2-} + H^+ \rightarrow Zn^{2+} + Cr^{3+} + H_2O$ (Acidic Medium)

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [5] Derive an expression for the radius of hydrogen atom by applying Bohr's atom theory.

(b) [4] How many moles are present in 7.5×10^{20} molecules of C_6H_6?

QIV.

(a) [5] Give the experimental verification of Graham's law of diffusion.

(b) [4] Draw molecular orbital diagram for oxygen molecule and also calculate its bond order.

QV.

(a) [5] What is Le-Chatlier Principle? Apply this principle to see the effect of change in pressure and temperature on gaseous equilibrium.

(b) [4] Define order of reaction. Give the examples of 1st, 2nd and 3rd order reactions.

QVI.

(a) [5] What is elevation of boiling point? Derive an equation which relates elevation of boiling point with molecular mass of solute.

(b) [4] Write short notes on: (i) Azimuthal Quantum Number (ii) Electrode potential